

PlantAL: An Interactive Active Learning System for Adaptive Tomato Disease Diagnosis

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Abstract

We present PlantAL, a browser-based interactive machine learning system for tomato disease classification and adaptive treatment recommendation. PlantAL combines an active learning loop, where users label the most uncertain images first, with two complementary treatment adaptation mechanisms: a Bayesian multi-armed bandit for offline deployment, and an LLM-powered module that generates personalized treatment recommendations from free-text farmer feedback. Both modules operate on top of a shared MobileNetV1 transfer learning backbone fine-tuned entirely in the browser via TensorFlow.js. The system requires no server infrastructure and is designed for agronomists and farmers with no machine learning expertise.

1. Introduction

Tomato cultivation is a critical agricultural activity worldwide, yet crop losses from fungal, bacterial, and viral diseases remain a major challenge for small-scale farmers. Early and accurate disease identification is the first step toward effective intervention, but access to expert agronomists is often limited in rural and low-resource settings.

Machine learning models trained on curated datasets such as PlantVillage [Hughes & Salathé, 2015] have demonstrated strong performance for disease classification under controlled conditions. However, these models often fail to generalize to real field conditions due to differences in lighting, camera quality, and geographic disease strains. Re-training such models for specific contexts requires labeled data that is expensive and slow to collect.

Interactive Machine Learning (IML) addresses this by placing the human expert in the training loop, allowing models to be adapted incrementally with targeted feedback [Fails & Olsen, 2003; Amershi et al., 2014]. PlantAL applies IML principles to the agricultural domain, combining active learning for model adaptation with two distinct interaction paradigms for treatment recommendation.

2. Scenario & Users

2.1 Context of Use

PlantAL targets two complementary use cases. In the first, an agricultural researcher or extension officer builds a disease classifier adapted to a specific region or farm. They upload a batch of unlabeled field photos, and the system guides them to label only the most informative images. In the second, a farmer consults the Field Assistant: they

photograph a diseased leaf, receive a diagnosis and treatment plan, apply it, and later report back on its effectiveness in their own words.

2.2 Users

The primary users are agricultural extension officers (medium technical expertise, high domain knowledge) and smallholder farmers (low technical expertise, high local knowledge of their specific conditions). Neither group requires knowledge of machine learning. The interface uses plain language throughout and never exposes model internals.

2.3 Active User Involvement

In the Label & Train module, users actively steer the model by deciding which images to label. The system presents the most uncertain image first, ensuring each label has maximum informational value, a core principle of pool-based active learning [Settles, 2009]. In the Field Assistant, users provide natural language feedback that an LLM incorporates into subsequent treatment recommendations, creating a continuous improvement loop driven entirely by field observations.

3. Application Design

3.1 Workflow

The application is structured around two tabs sharing a single classification model:

- **Label & Train:** The user uploads a batch of images. The system extracts MobileNet features from all images immediately, then presents the most uncertain image for labeling. Every 5 labels, the classifier retrains automatically. After retraining, the entire pool is rescored and reranked by uncertainty.

- **Field Assistant:** The user uploads a single field photo. The classifier identifies the disease and the treatment module generates recommendations. After applying the treatment, the farmer writes a free-text description of the outcome. This is stored and passed to the LLM as context on future uploads of the same disease.

3.2 Interface

The Label & Train tab uses a three-panel layout: a thumbnail pool on the left (images sorted by uncertainty with color-coded borders), a central labeling stage, and a right panel showing class coverage, accuracy history, and a training log. The Field Assistant tab uses a two-panel layout: the uploaded image on the left, disease diagnosis and treatment plan on the right, with a scrollable feedback history below.



Figure 1. Field Assistant tab showing diagnosis results and adaptive treatment recommendations after farmer feedback.

4. Implementation

4.1 Dataset

For the demonstration, a pool of 60 images was manually assembled from publicly available online sources, with 10 images per class across the six tomato disease categories. In production use, the pool would be constructed entirely from the farmer's own field photographs, eliminating the need for any pre-existing dataset.

4.2 Classification Model

The classification backbone is MobileNetV1 (0.25 width multiplier, 224×224 input) loaded from the TensorFlow.js model hub. The feature extractor is frozen; only the classification head is trainable. The

head consists of two Dense layers (128 and 64 units, ReLU activation, L2 regularization) with Dropout (0.4 and 0.3) and a softmax output over 6 classes. Training uses Adam (lr=0.001) with categorical cross-entropy loss. All computation runs in the browser via WebGL or CPU fallback, no server is required.

Uncertainty is measured as normalized entropy over the softmax output, used to rank unlabeled images for active learning query selection.

4.3 Treatment Recommendation Modules

Bandit version: Each treatment option is modeled as a Beta distribution with parameters α (successes) and β (failures). Thompson Sampling draws a random score for each treatment at query time and ranks accordingly. User feedback updates α or β : "worked" increments α for shown treatments, "failed" increments β , causing the system to explore alternatives in subsequent queries. This approach requires no internet connection.

LLM version: On each diagnosis, a structured prompt containing the disease name, its clinical description, and the full history of free-text farmer feedback for that disease is sent to the Groq API (Llama 3.1 8B). The LLM generates four actionable treatment steps adapted to the farmer's specific experience. Feedback accumulates per-disease, so the model progressively builds a contextual understanding of what works in that farmer's specific conditions.

4.4 Implementation State

The full active learning loop, both treatment modules, feedback history, accuracy tracking, and pool visualization are fully implemented and functional. The system currently supports single-session operation, treatment memory and labeled samples are not persisted across page reloads. Persistent storage and multi-user deployment are identified as future work items.

5. Evaluation

5.1 Proposed Method

We propose a within-subjects comparative evaluation between the active learning query strategy and a random sampling baseline. Participants would be agricultural extension officers recruited through a regional cooperative. Each participant would complete two labeling sessions: one guided by uncertainty sampling (PlantAL) and one with randomly ordered images. Both sessions would use the same pool of 60 images.

5.2 Metrics

- Label efficiency: classification accuracy as a function of number of labels provided (learning curve)
- Time to threshold: number of labels required to reach 80% validation accuracy
- Treatment adoption rate: percentage of LLM-generated recommendations the farmer reports trying (Field Assistant module)
- User workload: NASA-TLX questionnaire after each session
- Feedback quality: inter-rater agreement on the informativeness of farmer feedback texts

5.3 Hypothesis

We hypothesize that uncertainty-guided labeling will reach 80% accuracy with significantly fewer labels than random sampling (H1), and that LLM-generated treatment recommendations incorporating prior feedback will be rated as more relevant than static recommendations by domain experts (H2).

5.4 Preliminary Pilot Experiment

As a preliminary pilot, the active learning loop was run on the 60-image pool over 10 retrains totaling 62 labeled samples. Validation accuracy followed an expected learning curve: early retrains on small samples (5–10 images, train-only evaluation) showed high but unreliable accuracy due to overfitting, while validation accuracy (activated automatically once 20 labeled samples were reached) stabilized and trended upward from 43% at retrain 8 to 67% at retrain 10 across all six classes. Loss decreased consistently across epochs in every retrain session, confirming that the fine-tuning mechanism was learning rather than memorizing. One observable limitation was that labeling 9 consecutive Yellow Curl images in the final sessions introduced class imbalance, causing the model to plateau. This suggests that in a real evaluation, enforcing minimum class coverage constraints in the active learning query strategy would be beneficial, a refinement left for future work.

References

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